

Felix Sylvest Olesen

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[GitHub](#) | [LinkedIn](#) | [Portfolio](#)

PROFESSIONAL SUMMARY

Software engineer and Computer Science graduate currently completing an MSc in Computing (Software Engineering) at Imperial College London. Industry experience across full-stack development, QA automation, and shipped commercial software, including interactive educational products delivered under contract for universities and museums. Comfortable across the stack and the full software lifecycle — from React and C#/Unity front-ends to Python machine-learning systems, containerised deployment, and performance profiling — with a record of published research and self-directed projects.

SKILLS

Languages: Python, JavaScript, TypeScript, C#, C++, PHP, SQL, Ruby, Java, R

Frameworks & Libraries: React, Svelte, Vue.js, Flask, Ruby on Rails, Tailwind CSS, PyTorch, jQuery

Tools & DevOps: AWS, Docker, Kubernetes, Grafana, Jenkins, Jira, Git, Unity, Figma

Spoken Languages: English, Danish, Spanish (trilingual)

EDUCATION

MSc Computing (Software Engineering), Imperial College London

Sept 2025 – Sept 2026 (expected)

Thesis: Closing the Loop on Spec-Driven Development

- Software Engineering for Machine Learning Systems — trained and deployed an AKI-prediction model, hardened it against incomplete and faulty data, and kept it highly available using Docker, Kubernetes, and Grafana monitoring.
- System Performance Engineering — profiling and writing high-performance C++ using tools such as perf and Google Benchmark.
- Network & Web Security — network and browser exploitation techniques and defences, with hands-on attacks against VM-hosted sites.
- Scalable Systems & Data — state-of-the-art datacentre technologies and management, from BigTable to Spark.
- Software Engineering for Industry — microservices, cloud computing, software architecture, DevOps, sustainable software, and ML in production.

BSc (Hons) Computer Science with Sandwich Year, University of Surrey

July 2023

Grade Average: 79.6% (First Class Honours)

WORK EXPERIENCE

Software Developer (Contract), Education Evolved

April 2025 – October 2025

Startup building immersive educational experiences commissioned by universities and museums.

- Developed interactive digital exhibits and educational games using Unity, C#, and React, taking products from concept to shipped release.
- Shipped HELM, a multiplayer WebGL escape-room game that trains container-ship crews on key aspects of their roles.
- Built CEASEFIRE, a dialogue-driven negotiation game teaching young peacemakers the fundamentals of brokering real-world ceasefires.
- Created an interactive Bayeux Tapestry exhibit, letting users scroll the tapestry and explore individual elements to learn what each depicts.

Software Developer Intern, ICHK

December 2023 – February 2024

Open-source 'Gibbon' education platform for a Hong Kong secondary school; joined via prior Gibbon hackathon participation.

- Refactored legacy browser-alert code into a reusable, object-oriented alert module, improving maintainability across a large PHP codebase.
- Re-designed the library catalogue with a Netflix-style rails UI (Tailwind CSS) and a bookshelf-style backend (SQL), making it more engaging for students.
- Built a new house-points display for students and faculty, gathering requirements directly from end users to scope the feature.

Associate Test Engineer, Sky

June 2021 – July 2022

Industrial placement year on the SkyQ set-top-box platform.

- Performed end-to-end sanity and regression testing of video-on-demand services across all eight SkyQ machines.
- Built a full-stack test tool (Vue.js, Tailwind CSS, Flask) to push VOD assets to test machines, streamlining catalogue updates for the QA team.
- Automated test cases with an in-house Python framework, freeing time for higher-priority bug investigation and earning strong professional feedback.
- Coordinated across QA Automation, DevOps, and E2E testing teams to maintain tooling and reduce test-blocking obstacles.

PERSONAL & RESEARCH PROJECTS

AI Game Tester & Level Generator (BSc Dissertation)

Sept 2022 – June 2023

C#, Unity, ML-Agents, Python, Reinforcement Learning

- Built on EA SEED's ARLPCG research to create a reinforcement-learning agent that automatically tests games and generates levels, targeting consumer-grade hardware.
- Extended the design with Imitation Learning ("IL-ARLPCG"), seeding training from human gameplay to converge faster and reach higher reward than the original approach.

Genetically Modified Wolf Optimiser — Published Research (arXiv)Sept – Dec 2022

Python, PyTorch, NumPy | arXiv:2301.08950

- Co-developed a hybrid optimiser combining Grey Wolf Optimisation, Genetic Algorithms, and Stochastic Gradient Descent to train neural-network weights without relying solely on gradient descent.
- Achieved the highest grade in the cohort and published the work on arXiv with lecturer endorsement.

Skin Cancer Classifier (Team Research Project)

Jan – July 2023

Python, PyTorch

- Implemented an unsupervised image-classification pipeline based on Google Research's SimCLR for multi-class skin-cancer detection.
- Combined the model with team-generated synthetic data for class balancing, achieving a 10% improvement in test accuracy.

Portfolio Website

Oct – Nov 2024

Svelte, TypeScript, Tailwind CSS, Figma, Vercel

- Designed and built a personal portfolio from scratch, learning TypeScript and Svelte in the process.
- Prototyped the design in Figma first, enabling fast implementation and consistent styling throughout.

QUAD-RE — Perspective-Shifting Endless Runner

Mar – June 2023

C#, Unity

- Built an endless-runner to practise software design patterns in a games context.
- Applied state-machine, singleton, and strategy patterns to keep a growing codebase debuggable and maintainable.

Aquetect — Game Jam Builder/ManagerApr 2024

C#, Unity

- Solo-developed an isometric Aqueduct builder/manager for the 7-day 2024 Pixel Jam, hand-creating all pixel art.
- Delivered a polished playable demo under tight time constraints by keeping the core game loop simple.

AWARDS & RECOGNITIONS

2019 — University of Surrey Prestige Scholarship

2019 — ICHK Growth Mindset Scholarship B

HOBBIES & INTERESTS

- Hiking & Camping — completed the 100km MacLehose Trail in Hong Kong over 5 days, and solo-hiked Mount Fuji in a day.
- Game Development — always have a game jam or indie-game project on the go.
- Touch Rugby — played competitively in Hong Kong leagues, including against a side captained by a former world-number-one national team captain.